

Exercise Class 4: Growth and Limits to Growth in Early Industrial Capitalism

Political & Economic Thought — Master Notes · by Jakob V. Stangel

Readings: excerpts from J. S. Mill, *Principles of Political Economy* (1848) · Konrad Zweig, "Smith, Malthus, Ricardo, and Mill: the Forerunners of Limits to Growth" (*Futures*, Dec. 1979, pp. 510–23) · **Wikipedia**, *The Jevons Paradox* · Brett Clark & John B. Foster, "William Stanley Jevons and *The Coal Question*" (*Organization & Environment* 14(1), 2001, pp. 93–98) · **film:** *Snowpiercer* (2013).

The class's question. Building on Module 6, this class asks the big one: **is economic growth self-sustaining — inherent in capitalism — or does it hit limits?** The startling answer from **classical political economy** is that growth **ends**: every classical thinker (Smith, Malthus, Ricardo, Mill) expected the economy to grind toward a "**stationary state**." The class then follows the idea forward — **Mill** re-imagines the stationary state as *desirable*; **Zweig** casts the classicals as the **forerunners of the 1972 "Limits to Growth"** debate; and **Jevons** extends limits from *land/food* to *coal/energy*, giving us the **Jevons paradox** — the 19th-century ancestor of today's climate & "green-growth" argument.

★ **The red thread:** for classical political economy, growth is **NOT inherent or eternal** — it is a **transient phase** that runs into **natural limits** and halts at the **stationary state**. Two engines drive it to a stop: **diminishing returns** (Ricardo — poorer land, rising rent, falling profits) and **population** (Malthus — mouths outrun food). **Mill** agrees on the mechanics but flips the mood: the stationary state is a chance to stop the "**struggle for riches**" and live well. **Jevons** adds the modern twist: making resource use more *efficient* doesn't save the resource — it **burns more of it** (the **Jevons paradox**).

The whole class is the 1800s inventing "limits to growth" — 150 years before the climate debate.

Key concepts to apply: *the stationary state · diminishing returns · the falling rate of profit · the Malthusian population limit · Mill's production vs distribution · the "Art of Living" · limits to growth (Club of Rome 1972) · the Jevons paradox / rebound effect · the Coal Question · green growth.*

The classical growth story — why growth **ends** (recap of Module 6)

Core: all four classicals share one structure — an economy grows for a while, then **runs into a natural ceiling** and settles into the **stationary state**. They disagree on the *ceiling* and on how to feel about it.

Adam Smith — the optimist who still saw an end

Core: Smith is the founder of **growth economics** — but even *he* expected growth to stop eventually.

- **The engine of growth** (Module 3): the **division of labour** raises productivity → generates a surplus → **capital accumulation** → a **widening market** → *more* division of labour. A **virtuous circle** ("the

natural progress of opulence"). Growth is driven by **saving and investment** and by **free markets** (the invisible hand).

- **But growth is finite.** As capital accumulates, **competition drives the rate of profit down**. A country that had acquired its "**full complement of riches**" — the most its soil, climate and institutions allow — would reach a **stationary state**. Smith's three moods: the **progressive** state is "cheerful"; the **stationary** state is "dull"; the **declining** state is "melancholy." → *Even the apostle of growth built in a ceiling.* (Working Q1.)

What is "the stationary state"?

Core: the **end-point** of the classical growth story: an economy with **zero net growth** — capital and population no longer expanding, wages at subsistence, profits at their minimum, no further accumulation.

- Not a **crash** but a **plateau**: output stops rising, the population stops growing, and the "progress of wealth" halts. For Smith and Ricardo it is a **gloomy** destination (low wages, low profits, no dynamism); for **Mill** it is an **opportunity** (below). (Working Q2.)

Malthus & Ricardo — sustained growth is impossible

Core: the two "dismal economists" both deny that growth can go on forever — for **two different reasons**. (Full treatment in Module 6; here is the growth-specific version.)

	Malthus	Ricardo
The main problem	population outruns food (geometric vs arithmetic) → misery holds living standards at subsistence	diminishing returns on land → rising rent → falling profits → growth stalls (the stationary state)
Why growth can't last	any rise in wages just breeds more people, who eat the surplus → back to subsistence	as population grows, farming spreads to poorer land ; the landlord's rent rises and eats the capitalist's profit until investment stops
The solutions	moral restraint (later marriage) + abolish the poor laws ; unproductive spending by landlords cushions gluts	free trade — repeal the Corn Laws to cheapen food, lower rents, restore profits and <i>postpone</i> the stationary state

- Both agree on the **labour theory of value** and on **abolishing the poor laws**; they **disagree** on the **landlords** (Malthus: useful protectors; Ricardo: parasitic free-riders) and on **free trade** (Ricardo for; Malthus wary). (Working Qs 3–5.)
- **The shared verdict:** *growth is a temporary windfall, not a permanent feature of the market economy.* The market left alone runs **down** toward stasis. → This is the classical "limits to growth," and the pivot into **Mill** and **Jevons**.

John Stuart Mill — *Principles of Political Economy* (1848)

Reading: J. S. Mill, *Principles of Political Economy*, selections (the "Preliminary Remarks"). ($\triangle$ The excerpt is the opening survey — it delivers Mill's **definition of political economy** and the **production/distribution** distinction, but **not** "Of the Stationary State"; that famous chapter (Bk IV, ch. VI) is supplied below from the course + Zweig, clearly marked.)

Core: Mill is the **culmination** of classical political economy — he accepts the classical machinery but adds a **liberating twist** that turns the dismal science into a **reformist** programme.

What is political economy, for Mill?

Core: political economy is a **moral and social science** — the study of the causes of wealth that are **human**, not physical.

- Mill draws a line: in so far as a nation's economy turns on **physical knowledge**, it belongs to the **physical sciences**; but in so far as the causes are "**moral or psychological, dependent on institutions and social relations, or on the principles of human nature**," they are the object of **political economy**. *In plain terms:* economics studies the part of wealth that depends on **how humans organise themselves**, not on the laws of matter.
- It **assumes** the physical conditions of production (leaving them to physical science) and traces the "**secondary or derivative laws**" of the production and distribution of wealth. (Working Q6.)

★ The decisive break — laws of *production* vs laws of *distribution*

Core: Mill's single most important move, and the one that **splits him from Ricardo**. The laws of **production** are fixed by nature; the laws of **distribution** are **made by us** — and so can be **reformed**.

	Laws of Production	Laws of Distribution
Source	the properties of matter + the state of knowledge	" partly of human institution " — a society's <i>statutes and usages</i>
Fixed or chosen?	fixed / quasi-physical — "not an arbitrary thing"	chosen — society "has the power of deciding what institutions shall exist"
Reformable?	no — they constrain <i>how much</i> there is	yes — how the pie is <i>divided</i> is ours to decide

- The quotable line: "Unlike the laws of Production, those of Distribution are partly of human institution: since the manner in which wealth is distributed in any given society, depends on the statutes or usages therein obtaining."
- The guardrail (Mill is no wishful utopian): "though governments or nations have the power of deciding *what* institutions shall exist, **they cannot arbitrarily determine how those institutions shall work**" — once you choose a distributive scheme, its **consequences** follow law-governed patterns.
- Why it matters — this is where the dismal science stops being fatal. For **Ricardo**, distribution (rent, subsistence wages, the profit-squeeze) was fixed by **natural law**, so poverty was inevitable and

reform futile. Mill answers: **distribution is a human choice** → **redistribution and reform are legitimate and possible**, *without* violating any economic law. This is the intellectual charter of **social liberalism** — and why Mill grew sympathetic to a qualified **socialism**, **worker co-operatives**, and bettering the condition of the working class. (Working Q7.)

★ The stationary state — *welcomed*, not dreaded (Bk IV, ch. VI)

Core: (from Mill's famous chapter, beyond the excerpt — see Zweig.) Mill agrees with Smith/Ricardo that **growth ends** in a stationary state — but he **inverts the mood**. Where they saw gloom, Mill sees liberation.

- **Growth was never the point.** Mill is "**not charmed**" by the ideal of a life spent "**trampling, crushing, elbowing, and treading on each other's heels**" — the endless "**struggle for riches**." That competitive scramble is a *means*, useful only while poverty must be conquered — not a way to live.
- **A stationary economy ≠ a stationary humanity.** "A stationary condition of **capital and population** implies no stationary state of **human improvement**." With growth halted, energy is freed for the "**Art of Living**" — mental, moral and social cultivation, and **leisure** — rather than more *getting*.
- **Proto-environmentalism** (why the class reads him under "limits to growth"). Mill did **not** want every acre turned to human use: he dreaded a world with "**not a spot left where a wild shrub or flower could grow**," no solitude, no nature to contemplate. He hoped humanity would be "**content to be stationary, long before necessity compels them to it**." → *A 19th-century case for degrowth by choice* — *quality of life over quantity of output*. (Working Q8, in part.)
- **Where Mill lands vs his predecessors:** he **keeps** the classical framework (stadial history, Malthusian population as a real force, production as quasi-physical, the stationary state as the endpoint) but **breaks** on the two things that matter — **distribution is a human choice** and the **stationary state is desirable**. *The dismal science, turned hopeful*.

Zweig — the classicals as "forerunners of limits to growth"

Reading: Konrad Zweig, "Smith, Malthus, Ricardo, and Mill: the Forerunners of Limits to Growth" (*Futures*, 1979).

Core: Zweig's thesis pulls the whole class together — the four classicals, writing at the *dawn* of industrial capitalism, were the **original theorists of "limits to growth"**, anticipating the famous **1972 Club of Rome / MIT report *The Limits to Growth*** by 180 years. All four thought growth was **good but not permanent**; all four saw it ending in a **stationary state**.

- **Their common vision** (Zweig's spine): "**pressure of population, nature's niggardliness and decreasing response to human effort... subsistence wages, and falling profits**" → the economy ends "**in a state of precariousness and indigence**." **Smith, Malthus, Ricardo feared this end** as collapse; **only Mill welcomed it** — but Mill's benign version needs social reform the others never considered.
- **Why growth is NOT inherent in the market economy** (the exam pivot — **Working Q8**). The market is brilliant at **allocation** and at *generating growth while resources are ample* — but it has **no**

mechanism to overcome the finite limits of nature. The very success of accumulation triggers the sequence that undoes it:

accumulation → higher wages → population grows → more food demand → poorer land cultivated → *diminishing returns* → rising food cost + rent, wages pinned at subsistence → *falling rate of profit* → accumulation stops → *stationary state*. So growth is a **phase**, not a permanent property, of capitalism. Left alone, the market runs *toward* stasis. **What to do?** — postpone it (Smith: free trade, education; Ricardo: cheap food via free trade), restrain population (Malthus's moral check), or — the only real programme — **choose it and reform distribution** (Mill).

- **The modern parallel — and where the classicals differ:**
 - *The parallel:* the MIT "world model" of exponential growth hitting finite limits **"remarkably concurs"** with the classicals' mental models. Boulding's image: the **"cowboy economy"** of the open frontier becomes **"spaceship Earth"** — finite, closed, no escape.
 - *The differences:* the classicals' limit was **land / food** (agricultural diminishing returns + population); the moderns' is **energy, minerals & pollution** (industrial/ecological). And the classicals **under-weighted technical progress** — they thought innovation could only *postpone*, never *defeat*, diminishing returns (which is why the booming 19th-century economy seemed to *refute* Ricardo).
- **Zweig's own provocative conclusion** (*his advocacy, going beyond the classicals*): a genuine stationary state **cannot** be run by a free market — it would need a **"new interventionist order"** (central planning, controlled distribution, population control), because a no-growth economy can't satisfy everyone's rising claims through market forces. He ends on **Schumpeter** — "*Can capitalism survive? No, I do not think it can*" — and **Smith** — "*Life is hard in the stationary state, which is dull and melancholy.*"

★ William Stanley Jevons — *The Coal Question* (1865) & the Jevons paradox

Readings: Wikipedia, *The Jevons Paradox* • Clark & Foster, "William Stanley Jevons and *The Coal Question*" (2001). (*The second folder PDF is actually Missemmer's 2012 scholarly analysis of the same book, folded in below.*)

Core: Jevons **moves the limit to growth from LAND to ENERGY**. The classicals built limits out of food/land (diminishing returns); Jevons argues Britain's whole prosperity rests on a **finite, non-renewable fossil fuel — coal** — and that, paradoxically, using it *more efficiently* only makes us burn *more* of it. He is the 19th-century ancestor of every modern climate / "green-growth" debate.

▮ The Coal Question — the argument

Core: Britain's industrial supremacy is **coal-powered**, and coal is running toward exhaustion — not physically, but **economically**.

- **The chain:** Britain's dominance rests on **cheap, high-quality coal** → but coal is **finite & non-renewable**; as shallow seams are worked out, mining goes **deeper**, so the **cost of coal rises** → Britain **loses its cheap-energy advantage** → rivals with "virgin reserves" (USA) overtake it → **growth stalls**. A **coal-based stationary state** — the energy version of Ricardo's land-based one.

- **Malthus, transposed from food to fuel.** Jevons swaps corn for coal: "**the momentous repeal of the Corn Laws throws us from corn upon coal.**" Free trade meant Britain imported food and paid with **coal-made manufactures** — so the ultimate constraint became **energy, not land**. He puts *depletion* into an analytical frame just as Malthus did for population.
- **"Economic," not physical, exhaustion.** The mines are "**literally inexhaustible**" — we'll never scrape the bottom — but we'll "**have to pay dear for fuel.**" What binds is **cost**, not emptiness; and what matters is "**the rate at which our consumption increases**" (which is **exponential**), not the exact size of the reserve.

★ The Jevons paradox — efficiency *increases* consumption

Core: the idea Jevons is famous for. **Making a resource more efficient to use does NOT save it — it makes us use *more*.**

- **The paradox (the quote to memorise):** "It is wholly a confusion of ideas to suppose that the economical use of fuel is equivalent to a diminished consumption. The very contrary is the truth." *"It is the very economy of its use which leads to its extensive consumption."*
- **The mechanism** (why efficiency backfires): a more efficient engine **lowers the effective cost** of power → **profits rise, capital flows in, output prices fall, demand expands, and new uses open up** → and the boom **spreads to other industries**. So total demand for coal grows **faster** than efficiency saves it per unit. Efficiency is an **engine of expansion**, not conservation.
 - **Watt's steam engine (the proof):** each improvement made steam power **cheaper**, so it spread into ever more uses — every efficiency gain "**does but accelerate anew the consumption of coal.**"
 - **Blast furnace (in one sentence):** use less coal per ton of iron → "**profits increase, new capital is attracted, the price of pig-iron falls, but demand for it increases; and eventually the greater number of furnaces will more than make up for the diminished consumption of each.**"
- **The modern name — rebound & backfire:** the **rebound effect** = efficiency savings are **partly offset** because cheaper energy stimulates more use (a more efficient car → you **drive more**; plus the money saved gets spent on *other* energy-using things — and cheaper energy grows the *whole* economy). When rebound **exceeds 100%**, total use **rises** — that's "**backfire**" = **the Jevons paradox proper**. (*Jevons anticipated not just the direct rebound but the indirect and macro forms — remarkably complete.*)

Why it matters — the challenge to "green growth"

Core: the paradox is the **foundational objection to techno-optimism**. If **efficiency alone** can't cut resource use, then **technology cannot decouple growth from its environmental cost** — you need **caps, taxes, or structural change**, not just better gadgets.

- **Efficiency ≠ conservation** — the whole lesson in three words. LED bulbs, efficient engines, efficient data-centres: each *can* raise total energy use by making energy services cheaper and the economy bigger. → *A direct hit on "green growth" / decoupling.*
- **Jevons's own blind spot — substitution pessimism (his big mistake).** He examined and **dismissed every alternative** to coal: wind/water (unreliable), wood (already abandoned for coal), oil (needs coal to extract), **electricity** (not a *source* but a *carrier* — trusting it is like believing in "**perpetual motion**"), solar (not yet mastered; Britain's not sunny). He'd even *documented* an earlier

fuel switch (charcoal → coal in the iron trade) yet **couldn't imagine another**. He badly under-rated **petroleum and hydro** — Keynes later called the argument "over-strained."

- **His grim conclusion — a choice, not a fix:** "We have to make the momentous choice between brief but true greatness and longer continued mediocrity" — and he effectively chose to **burn it fast**. His one constructive proposal was **intergenerational compensation**: use the coal intensively but convert the wealth into **lasting assets** (education, infrastructure) and **pay down the National Debt** so posterity isn't doubly punished. (*He rejected a coal tax — because coal underlies everything, taxing it would tax the whole economy — an argument that eerily prefigures modern objections to a carbon tax.*)
- **Clark & Foster's ecological-Marxist reading:** Jevons spotted the *mechanism* but missed its *cause*. As an early marginalist he treated capitalism as natural, so he located rising demand in **individual behaviour + population** — never in **capital accumulation** itself, the "**treadmill of production**" that compels endless reinvestment and growth. (Their fix — the one that "never entered Jevons's mind" — is to **change the social relations of production**, not just the gadgets.)
- **The exam-worthy irony:** Jevons was a **pro-growth, pro-industry marginalist** who cared about **British power, not the planet** (he ignored coal's pollution and miners' deaths) — yet his logic became **environmentalism's sharpest weapon against unlimited growth**. The booster of growth handed the **degrowth** movement its best argument.

Snowpiercer (2013) — the case on film

Core: Bong Joon-ho's dystopia as a **parable of growth, limits, and distribution**. A botched climate geo-engineering fix freezes the Earth; the last survivors circle the dead planet on a perpetual-motion **train**, rigidly divided **tail (the poor) vs front (the rich)**.

- **How it ties to the class:** it dramatises the two threads at once — a **hard ecological limit** (a ruined planet; the train as a closed, finite system with a fixed "**ecological balance**" the elite police by culling the poor) **and** the **distribution** question (Mill's point that who-gets-what is a matter of **human institution**, here a brutal engineered hierarchy). → *A vivid prompt: is the answer to limits a better-managed hierarchy (the Engine), or breaking the system itself (the ending)? — exactly the Jevons vs Clark-&-Foster argument in fiction.*

★ Working-question answers

The class's eight working questions, worked. (1–5 recap Module 6; 6–8 are the new material.)

1 • **What was Adam Smith's view on economic growth?** *Answer:* see Smith on growth. Smith is the great **optimist**: growth comes from the **division of labour** → higher productivity → a surplus → **capital accumulation** → a **widening market** → still more division of labour — an **upward spiral** ("the natural progress of opulence"), steered by the **invisible hand** of self-interest and free competition, needing no state direction. *But* even Smith saw growth as **finite**: as capital piles up, competition drives the **rate of profit down**, and a country reaching its "**full complement of riches**" sinks into a **stationary state** — "**dull and melancholy**," with wages at subsistence. *It is growth itself, not the level of wealth, that benefits workers* (the "progressive state" is the happy one).

2 • What is "the stationary state/society"? *Answer:* see the stationary state. The end-point of the classical growth story: an economy with **zero net growth** — capital and population no longer expanding, wages at subsistence, profits at their minimum, no further accumulation. Not a crash but a plateau. For Smith and Ricardo it is a gloomy destination; for Mill a chance to stop the "struggle for riches" and live well.

3 • Malthus's & Ricardo's view on *sustained* growth? *Answer:* see Malthus & Ricardo on growth. Both say sustained growth is impossible — the economy must end in the stationary state — for different reasons: Malthus because population always catches up with any rise in output (mouths outrun food); Ricardo because diminishing returns on land raise rent and crush profits until accumulation stops.

4 • The main problem — Malthus? Ricardo? *Answer:* Malthus: the population principle — population grows geometrically while food grows only arithmetically, so living standards are pinned at subsistence by the checks (misery, vice, moral restraint). Ricardo: diminishing returns → rising rent → the falling rate of profit — as growth pushes farming onto poorer land, the landlord's rent rises and squeezes the capitalist's profit to a minimum, ending investment. (*Malthus blames nature/population; Ricardo blames the distribution of income — specifically the landlord.*)

5 • The solutions — Malthus? Ricardo? *Answer:* see Module 6 study answers. Malthus: moral restraint (later marriage) + abolish the poor laws (relief is counterproductive); landlords' "unproductive" spending cushions gluts. Ricardo: free trade — repeal the Corn Laws to cheapen food, lower rents, restore profits, and postpone the stationary state. Neither offers a true escape — only a delay.

6 • What is political economy according to Mill? *Answer:* see what is political economy. A moral and social science — it studies the causes of a nation's wealth that are human ("moral or psychological, dependent on institutions and social relations, or on the principles of human nature"), assuming the physical conditions of production (which it leaves to physical science). Its job is to trace the "secondary or derivative laws" of the production and distribution of wealth.

7 • How does Mill's programme differ from his predecessors? Where do they agree? *Answer:* see the production/distribution break and the stationary state welcomed. Agreement: Mill keeps the whole classical machinery — Malthusian population as a real force, production as quasi-physical, and the stationary state as the inevitable endpoint. The breaks (two, both decisive): (a) the laws of production are fixed by nature, but the laws of distribution are "partly of human institution" — chosen by society, so redistribution and reform are legitimate and possible (against Ricardo's fatalism that poverty is fixed by natural law); (b) the stationary state is *desirable, not dreaded* — a chance to end the "trampling, crushing, elbowing" struggle for riches, cultivate the "Art of Living," and preserve nature. → Mill turns the dismal science hopeful and reformist.

8 • Is economic growth inherent in capitalism? Why not? What to do? *Answer:* see why growth is not inherent. No — for classical political economy growth is a transient phase, not a permanent feature of the market economy. Why not: the market can allocate and generate growth while resources are ample, but it has no mechanism to beat the finite limits of nature; the very success of accumulation sets off population growth → diminishing returns → rising rent → falling profit → the stationary state. What to do: postpone it (free trade, education), restrain population (Malthus), or — the only real programme — choose the stationary state and reform distribution so it is comfortable rather than "pinched" (Mill). And Jevons adds the sting: you can't efficiency-your-way out — saving a resource per unit just makes you use more of it.

★ Fun facts & memorable details

Sticky bits from Exercise Class 4 (growth • the stationary state • Mill • Jevons).

- **The founders of economics were growth pessimists.** Every classical — even Smith — expected growth to **run out** and end in a "**dull and melancholy**" stationary state. Endless growth is a *modern* faith they didn't share.
 - **Mill wanted degrowth — in 1848.** He hoped humanity would "**be content to be stationary, long before necessity compels them to it,**" and dreaded a world with "**not a spot left where a wild shrub or flower could grow.**" A Victorian environmentalist a century before the word existed.
 - "**Content to be stationary**" ≠ **stagnant.** Mill's twist: "a stationary condition of **capital and population** implies no stationary state of **human improvement**" — stop *getting*, start *living* (the "Art of Living").
 - **The Jevons paradox: efficiency backfires.** "**It is wholly a confusion of ideas to suppose that the economical use of fuel is equivalent to a diminished consumption. The very contrary is the truth.**" Watt's *more efficient* steam engine led to *far more* coal burned — the 1865 ancestor of every "green-growth" debate.
 - **Jevons's own blind spot.** He dismissed **electricity** as an energy source — trusting it, he said, was like believing in "**perpetual motion**" — and badly under-rated oil. The prophet of energy limits couldn't imagine the energy transition.
 - "**Brief greatness or longer mediocrity.**" Jevons's grim choice for coal-burning Britain — and he effectively chose to **burn it fast.** Yet this pro-growth, pro-empire economist handed the **degrowth** movement its sharpest weapon.
 - **From "cowboy economy" to "spaceship Earth"** — Boulding's image (via Zweig) for the whole shift the classicals foresaw: from the open frontier to a finite, closed system with nowhere left to expand.
 - **Snowpiercer is a stationary-state parable on rails** — a closed, finite system where the elite police a fixed "ecological balance" by culling the poor. The question the film (and the class) leaves you with: *manage the hierarchy, or break the train?*
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